

Daily Review

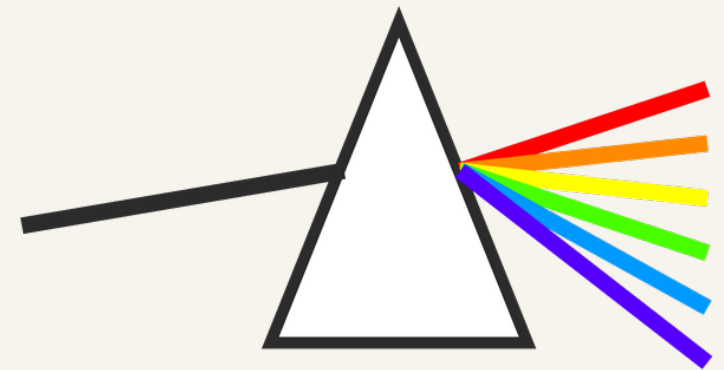
Booklet p.20-21 — The Big Bang, and p.5 — $W = mg$

- 1 State the estimated age of the universe.
About 13.8 billion years
- 2 Describe the universe at the Big Bang.
Very small, very hot, very dense
- 3 Name one piece of evidence for the Big Bang.
The microwave background
- 4 A 60 kg astronaut is on the Moon. Find their weight.
96 N
- 5 What is redshift?
Stretched light

SPACE

Spectra

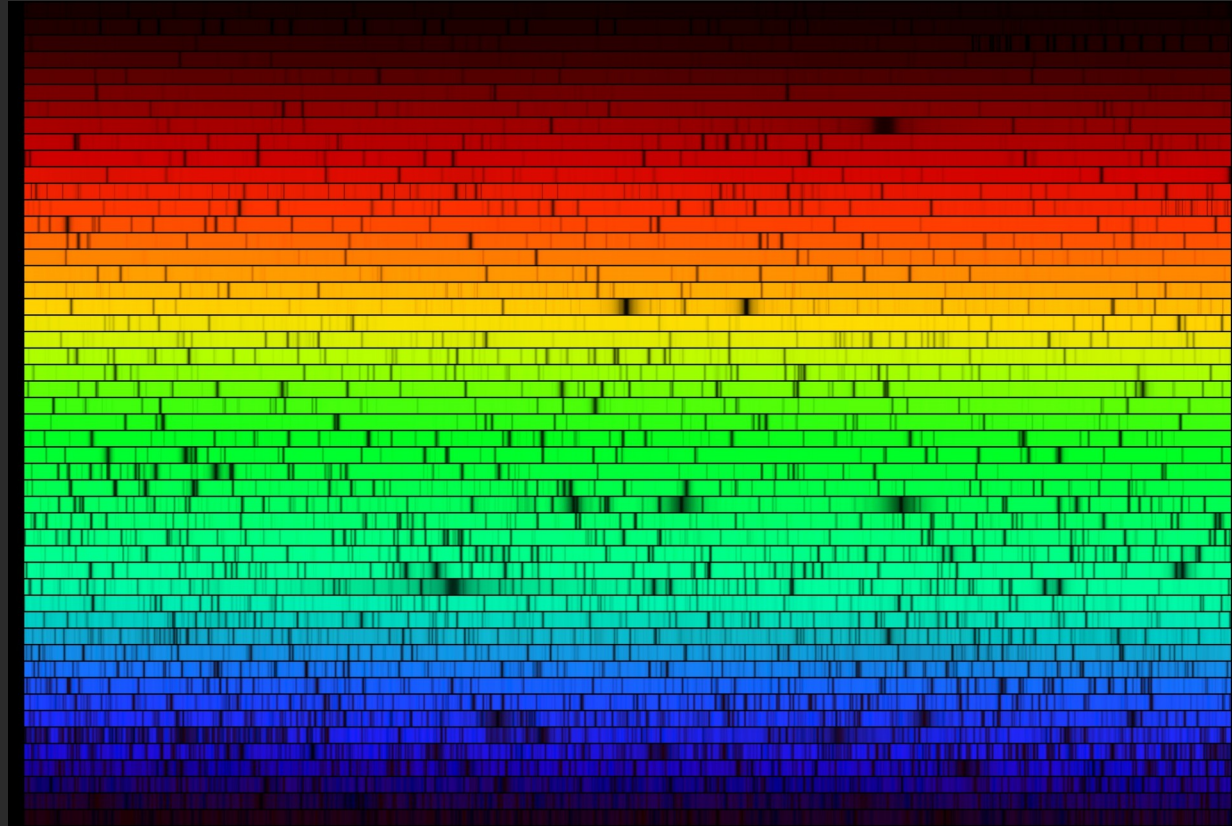
Every element leaves its own barcode in light.



National 5 Physics (S3)

THE BIG QUESTION

Nobody has ever been to a star. So how can anyone know what one is made of?



N.A. Sharp/KPNO/NOIRLab/NSO/NSF/AURA (CC BY 4.0)

PART 1

Spectra

Reading a star's spectrum to find out what it is made of.

By the end of this part...

- 1 I can identify continuous and line spectra
- 2 I can use spectral data to identify the elements in a star

Show me — Describe each word

SHOW ME BOARDS

1 spectrum

Light split into its
colours

2 continuous spectrum

Every colour, with no
gaps

3 line spectrum

Separate bright lines
only

4 element

A substance made of one
kind of atom

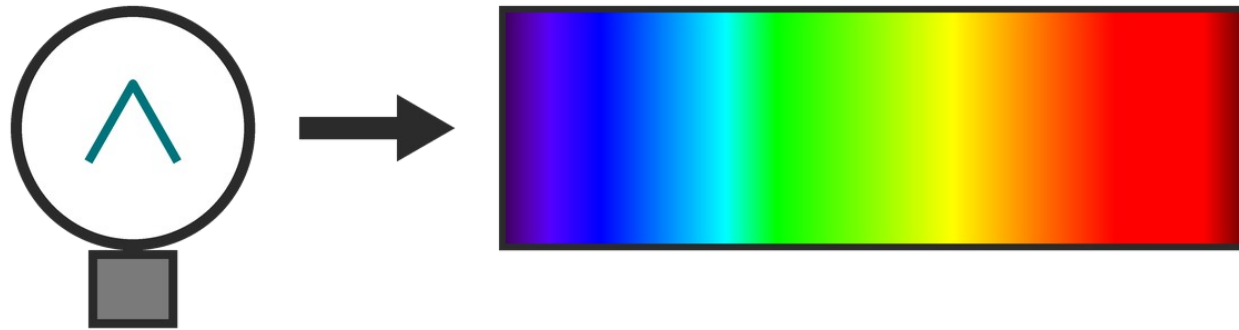
5 spectroscope

The instrument that
splits light

A continuous spectrum

Booklet p.22

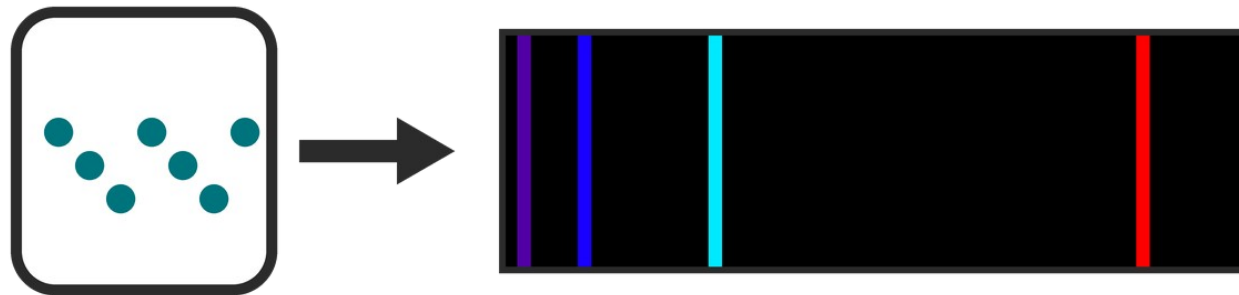
- A hot solid, like a lamp filament, gives out white light.
- Split it up and you see every colour, with no gaps.
- That is a continuous spectrum.



A line spectrum

Booklet p.22

- A hot gas of one element gives out only certain colours.
- Split it up and you see separate bright lines.
- Each element has its own pattern of lines — like a barcode.



Hinge question

FINGER VOTE

Think first — then show A to E on your fingers, 1 to 5

A glowing gas of one element is viewed through a spectroscope. What is seen?

- A** Every colour, with no gaps
- B** A few separate bright lines
- C** No light at all
- D** One single white line
- E** Every colour, plus extra lines.

Answer B

Show me – which spectrum?

SHOW ME BOARDS

EXTRA PRACTICE

- 1 Hydrogen gas glows. What type of spectrum?
- 2 How is a line spectrum different?
- 3 Which type of spectrum identifies an element?
- 4 Does a line spectrum have gaps?
- 5 What does a spectroscope do to light?

A line spectrum

Separate lines, not all colours

A line spectrum

Yes

It splits it up

Task 1 – the spectra experiment

Booklet p.22

Record each element's spectrum in colour.
Then work out what is in the mystery mix.

Booklet p.22, Task 1

Task 2 – draw both types

Booklet p.22

Use colour pencils to show a continuous spectrum and a line spectrum.

Booklet p.22, Task 2

Demo — see the lines yourself

DEMO

Look at a filament lamp through the glasses.
Then at a discharge tube, and at a street lamp.

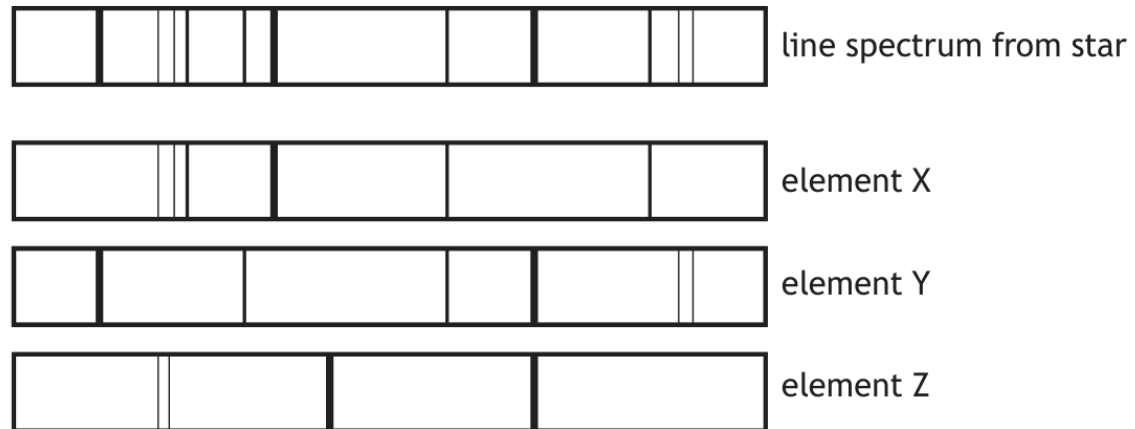
Predict first: which will show separate lines?
Sketch what you see beside your Task 2 drawings.

Reading a star's spectrum

Booklet p.22

- Split the star's light into a line spectrum.
- Line up each known element underneath it.
- Every line of an element is in the star: it is present.
- Even one of its lines missing: it is not present.

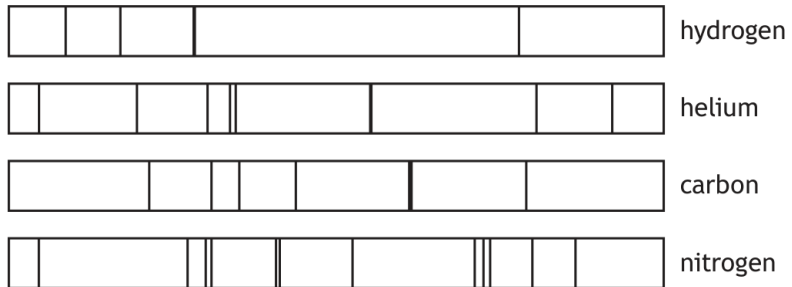
Worked example



- 1 Element X: all its lines are in the star.
- 2 Element Y: all its lines are in the star.
- 3 Element Z: one of its lines is not in the star.
- 4 Present: X and Y only.

Show me — what is in the star?

SHOW ME BOARDS



1

Is hydrogen present in the star?

No

2

Is helium present?

Yes

3

Is carbon present?

Yes

4

Is nitrogen present?

No

Show me — reading spectra

SHOW ME BOARDS

EXTRA PRACTICE

- 1 Two of an element's three lines appear.
Is it present?
No
- 2 What must be true for an element to
be present?
Every one of its lines is
there
- 3 Why compare a star with known
elements?
To identify them
- 4 Can a star contain more than one
element?
Yes
- 5 A star's line matches no element given.
So?
Another element is
there

Correct your booklet

Booklet p.22, Task 3

CROSS OUT AND FIX

Cross out “the star bellow”.
Write: “the star below”.

Task 3 – the example question

Booklet p.22

Use the spectral data to find which elements are present in the star.

Booklet p.22, Task 3

Task 3 – Answers

ANSWERS

Booklet p.22

- Helium and hydrogen are present – every one of their lines is in the star.
- Calcium and sodium are not – each has a line the star does not.

Checkpoint

- 1 What type of spectrum does a filament lamp give?
- 2 How is a line spectrum different from a continuous one?
- 3 How do astronomers tell which elements are in a star?

Continuous

Separate lines, not every colour

They match the lines to known elements

Exam practice

Past paper

- National 5 Physics 2024, Section 2, question 6(c). 1 mark.
- 2023, Section 2, question 5(b). 1 mark. Elements in a star.
- Then booklet p.23, Task 4.

LOOKING AHEAD

Light Years

Space is so big we measure it in how far light travels.

Plenary

- 1 The big question: how do we know what a star is made of?
- 2 Name the two types of spectrum.
- 3 Which type does a filament lamp give?
- 4 Which type does a glowing gas of one element give?
- 5 What instrument splits starlight up?

By reading the lines in its light

Continuous and line

Continuous

A line spectrum

A spectroscope

Plenary

EXTRA PRACTICE

- 1 Describe a continuous spectrum.
- 2 Describe a line spectrum.
- 3 Why is a line spectrum like a barcode?
- 4 One line is missing from the star. Is it present?
- 5 Why can we not collect a sample of a star?

Every colour, no gaps

Separate bright lines

It belongs to one element

No

Far too far away

Extension

- 1 Suggest why a mixture of gases gives more lines than one gas.
- 2 Two stars have identical line patterns. What can you conclude?
- 3 Why is spectroscopy the only way to find what distant stars contain?

Extension

EXTRA PRACTICE

- 1 Suggest how helium was first discovered in the Sun, not on Earth.
- 2 Why does a hotter filament still give a continuous spectrum?
- 3 Suggest a problem if the star's spectrum is very faint.

Course specification

N5 Physics course specification v4.0, p.8 – Cosmology

- Identification of continuous and line spectra.
- Use of spectral data for known elements, to identify the elements present in stars.